Modern Materials/chemicals.

Modern materials refers to the "materials with properties that traditional materials do not possess.

The chemical or physical properties of these materials can be changed by external factors like temp, light and electrical charges, so as to meet specific requirements in modern technologies.

Some modern materials in Chemistry include.

1) Composite materials:

These are made from two or more materials that are different in some way such as physically, chemically or biologically.

2) Nanomelerials They have unique optical, electrical, thermal and magnetic properties.

3) Polymers

These are man made polymers have been around mid 1800's Scientists are always working to create new Polymers with better physical properties

4) Semi Conductors

Solid State materials that produce or tarnsmit an electric current when a voltage is applied.. They are a key component of modern technology and electronic equipment.

5) Ceramics

Ceramics have been used for thousands of years but their applications have expanded greatly in modern times.

6) Liquid Crystals

These have some positional and orientational order but also behave as fluids. They are used in displays ranging from finger nail sized to wide screen televisions.

Ceramics

Defination: A ceramic is an inorganic non-metallic solid made up of clay that have been shaped and then hardened by heating to high temperature.

- Ceramic comes from the word meaning Pottery
- Ceramics have many applications other than pottery.
- Now a days ceramic includes materials like glass, advanced ceramics and some cement systems etc.

Properties of Ceramics.

The properties of ceramic materials are dictated by the types of raw materials present.

In general, most ceramics have following properties.

1: Ceramics are hard, extremely strong, brittle and having little elasticity.

2: Ceramics are corrosion-resistant, oxidation resistant and durable

3: Ceramics are refractory material with high melting point that is used in lining of inside walls of a furnace.

4: Ceramics are inert to chemical action and generally donot react with most liquids, gases, alkalies and acids.

5: Ceramics are thermal insulators and electrical insulators.

6: Certain ceramics conduct electricity such as chromium dioxide conducts electricity as most material do

7: Ceramics are non-magnetic but ceramics containing iron oxide can have magnetic properties. Iron oxide-based ceramics are called ferrites

8: Ceramics are generally oxides, carbides, nitrides, borides and silicates

Applications

Ceramics have application in the following areas.

1: Aero space

Ceramics are used in the formation of parts of space shuttle, rockets and space stations

2: Consumer Usage

Ceramics have great usage in our homes like glass ware, pottery, dinnerware and home electronics.

3: Automotive industry

For example catalytic converters, filers rotors, valves, spark plugs, thermostats, piston, rings etc are ceramics.

4: Medical (BioCeramics)

Ceramics are used in medical field especially in dental, and bone fillings and in bone implants

5: Military:

Structural components for ground, air and naval vehicles of military are made from ceramics

6: Computers and electronics:

Computer parts like insulators, resistors, superconductors capacitors etc are made up of ceramics

7: Abrasives

Abrasives are used for grinding, cutting, polishing of materials

- natural abrasives are garnet, dimond
- synthetic abrasives are silicon carbide diamond -fused alumina

8: Building and Construction

Manufacturers use ceramics to make bricks, tiles, piping, and other construction materials

9: Coatings

As ceramic materials are more corrosion-resistance than most metals, manufacturers often coat metal with ceramic enamel

Classification of Ceramics.

Ceramics can be classified into three distinct material categories

1: Oxides: Alumina, Zirconia, iron oxide etc

2: Non-Oxide: Carbides, borides, nitrides and Silicides

3: Composites: Combination of oxides and non oxides.

Plastics

The word Plastic is derived from Latin word **plastics** meaning capable of molding.

What is Plastic? Plastic is made up of large organic molecules that can be formed into a variety of products.

The molecules of plastics are long carbon chains called polymers, which give plastics many of their useful properties

Properties of Plastics

1: **Lighter**: They are lighter than many materials of comparable strength like metals and wood

Plastics have a lower density than that of metals so they ar Lighter.

2: **Durable**. Plastics do not rust or rot.

• They are Corrosion resistant or oxidation resistant.

3: **Variety of colours** Most plastics can be produced in any colour. They can be manufactured as transparent, translucent and opaque

4: **Mixed with other meterials**

Plastics can be strenghthened with glass and other fibers to form incredibly strong materials.

5: **Temperature tolerant**

Certains plastics are specifically designed to withstand temperaturs as high as 288°C. However general plastics are not used when high heat resistance is required.

6: **Insulator**: Plastics are electrical insulator.

7: **Non Magnetic**: Plastics are non magnetic in nature

8: **Chemically stable**: Plastics are chemically stable and some what inert.

9: **Non ductile & Brittle** Plastics are not brittle and ductile

Chemistry of Plastics

Plastic consist of a very long molecules each composed of carbon atoms linked into chains.
e.g Polyethylene plastics composed of long molecules that each contain over 200,000 carbon atoms
This long chain like molecules give plastics unique properties. This long chain structure distinguish plastics from material such as metal that have short crystalline molecular structure
Some plastics are made from plant oils but the majority are made from fossil fuels.
e.g Poly urethane - made from olive oil and linseed oil, used in foam insulation ~~panels~~ panels and sealants.
Coca cola's PET bottle made from 30% biomass

Types of Plastics

All Plastics can be divided into two main groups on basis of their response to heat.
1: Thermo splastics
2: Thermosetting Plastics.
These terms refer to different ways of response, these plastics show to heat due to their chemical structure of plastics.

Thermo Plastics

Thermoplastics can be softened and hardened repeatedly after being heated and coold ~~twice~~
For this reason thermoplastics can be remolded and reused

Thermo setting Plastics

1: Thermosetting plastics harden Permanently after being heated once.

2: Thermoplastic molecules are linear or slightly branched and are not chemically bonded with each other, when heated.

3: Thermoplastic chains are held together by weak Vander Waal forces that cause the long molecular chains to cluster together like piles of entangled Spaghetti

4: As thermoplastic materials consist of indiviual molecules, properties of thermoplastics are largely influenced by molecular weight.
e.g by increasing the molecular weight of thermoplastics the tensile strength, impact strength and fatigue strength is increased.

Examples:
Polyvinyl, Polystyrene, Polyethylene are examples of thermoplastics.

2: These plastics consist of chain molecules and chemically bonded or cross linked with each other when heated

3: When thermosetting plastics are cross linked, the molecules create a permanent, three dimensional network that can be considered one giant molecules

4: Many properties of thermo setting plastics are determined by adding different types and amount of fillers such as glass fibers

5: These plastics consist of a single molecular network, molecular weight does not significantly influence their properties.
These plastics are often used to make heat resistant products because these plastics can be heated to the temp of 260°C without melting

Examples. Thermo setting resin plastic Duraplast, melamine epoxy resin, acrylic.

Thermoplastics.	Thermosetting Plastics
Thermoplastics can be easily softened by heating and hardened by cooling.	Thermosetting plastics are hardened permanently after being heated once
Thermoplastics molecules are not or less branched	Thermosetting plastics have branched and complex structure
They behave like wax	They donot behave like wax
They are not brittle	They are brittle
e.g. Polyvinyl, polystyrene, Polyethylene.	e.g. Bakelite, duraplast, melamine, epoxy resin

Importance and Usage of Plastics

① Beddings: (a) They are used in pillows and mattresses. These beddings are filled with polyurethane or polyester.

(b) Blankets and bed spreads are made up of acrylic plastics and used in nylon carpets.

② Daily life use: (a) The cars we drive, computers, cooking utensils and recreational material like different games, houses and builidings includes important plastic components.

The average car contains almost 136 kg (12% of total weight) of plastic

(b) Telephones, textiles, compact discs, paints, plumbing fixtures, boats, furniture and other domestic products are made of plastics.

③ Aerospace Industry: The aerospace industry uses plastics to make strategic military parts for missiles, rockets and air crafts.

④ Key Industry: Plastics are used extensively by many key industries including the automobile, aerospace, construction, packaging and electrical industries

⑤ Plastics are also used in specialized fields such as health industry to make medical instruments, dental fillings, optical lenses and biocompatible joints.

Advantages of Plastics

1) Plastics are light in weight and can be easily moulded.
2) Plastics possess very good strength and toughness
3) Plastics possess good shock absorption capacity.
4) Plastics are corrosion-resistant and chemically inert.
5) Plastics possess good thermal and electrical insulating property.
6) Plastics are very good water resistant and possess good adhesiveness
7) Plastics are a recyclable material and doesnot decompose

Disadvantages of Plastics

1) Plastic is a non-renewable resources
2) Plastics are brittle at low temperature
3) Plastics deform under load
4) Plastics are low heat resistant and have poor ductility
5) Plastics are combustible
6) Their recycling process is very costly

Impact on Environment

Hazards of Plastics

1. Although durability is a +ve characteristics of plastics but it is actually highly harmful for the environment as it is slowly degradable
2. Chemicals added to plastics are absorbed by human bodies, Some of these chemicals found to alter hormones and other harmful effects.
3. Plastic debris, mixed with chemicals often swallowed by marine animals and are poison for wildlife
4. Floating plastics serve as dangerous materials for environment
5. Plastics buried deep in landfills can penetrate harmful chemicals that spread into ground water
6. About 4% of oil production is used as raw material to make Plastics Similar amount is consumed as energy in the process
7. Burning plastics can sometimes result in toxic fumes

Some times plastics help the environment in several ways
As plastics has been used to make cars lighter as a result, less oils is used to mobilize the cars and less CO_2 is emitted.

Plastic Waste mangment

Different methods for reducing and disposing of plastic wastes are being explored. Some options are
1) Reducing consumption of plastics using biodegradable plastics in incinerating or recycling plastic waste
This strategy is called **3R — Reduce, reuse and recycle**

1) Source Reduction:

- Source reduction is the practice of using less material to manufacture a product. e.g wall thickness of many plastic has been reduced in recent years.
- Some European countries have proposed to eliminate packaging that cannot be easily recycled.

2) Biodegradable Plastics

- Due to their molecular stability, plastics donot easily break down into simpler components so plastics are not considered biodegradable.

Researchers are working to develop biodegradable plastics that will disintegrate due to bacterial action or exposure to sunlight
e.g: 1 Scientists are incorporating starch molecules into some plastic resins
2 when these plastics are discarded bacteria eat the starch molecules this cause polymer molecules to break apart allowing the plastic to decompose.

3). Incineration:

- Some wastes such as paper, plastics wood and other flammable materials can be burned in incinerators. Resulting ash requires much less space for disposal than original waste.

As incineration of plastics can produce hazardous air emissions and other pollutants so this process is strictly regulated.

4) Recycling Plastic

All Plastics can be recycled

- Thermoplastics can be remelted and made into new products
- Thermosetting plastics can be ground, mixed and then used as filler in moldable thermoplastic materials.
- Highly filled and reinforced thermosetting plastics can be crushed and used in new composite formulations.
- It takes around 100 to 500 years to get the plastic degraded naturally that's why concept of plastic recycling is promoted

Types of Plastics

There is no universal standard to classify plastic

In 1988 The Society of plastics Industry (SPI) designed the symbol code for plastic

- The symbol code consists of a triangle of arrow inside which they are chambers ranging from 1-7
- Most of us are familiar with this symbol embossed on plastic items like Pet bottles, Yogurt cups etc.

Following are the types of Plastics

1: Plastic Number 1 (PET)

Polyethylene Terephthalate also known as PETE or PET

Used in disposable Soda bottles, medicine containers and water bottles making.

2: Plastic number 2 (HDPE)

High-density Polyethylene

Used in making most milk jugs, detergent bottles, Juice bottles butter tubs and toiletries bottles.

3: Plastic number 3 (PVC) Poly vinyl chloride

It is used to make food wrap, shower curtains, windows and door frames, plastic pipes, automotive parts, bottles for cooking oil and highly common plumbing pipes

4: Plastic number 4 (LDPE) Low density Polyethylene

It is used to make grocery bags, some wrapping films, squeezable bottles and bread bangs.

5: Plastic number 5 (PP) Poly Propylene

It is used to make yogurt cups, medicine bottles, ketchup, syrup bottles, straws, wide-necked containers, microwave-able bowls and bottle caps

6) Plastic Number 6 Poly propylene.

It is used to make yogurt cups, bottles etc.

7: Plastic Number 7

This category basically means everything else. Product produced include baby and water bottles, sports equipments medical and dental devices, CD's, DVD's and even iPods.

- Plastic number, 1, 2 and 6 are commonly collected for recycling but 3, 4, 5 are not recycled in a regular matter.
- Plastic items without number are also not fit for recycling

Plastics in categories 2, 4 and 5 are generally considered safe

However anyother categories should be used with extreme caution